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E-102 ISOLATED PHASE BUS DUCT

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of all Isolated Phase Bus Duct in accordance with Part 4, Subsection 1.2 “Summary of Work”.

One (1) complete set of isolated phase bus duct shall be provided for steam turbine generator output units.

Each generator shall be provided with an isolated-phase, metal enclosed bus duct system. This shall include lengths of three-phase bus duct, complete with tee-offs and connections for the generator, generator transformer, unit auxiliary transformer, generator circuit breaker, voltage transformer, surge protection cubicle, connections for the static excitation transformer (if required) and static frequency converter (if required).

The above shall be complete including all accessories and appurtenances required for a completely functional system and as hereinafter specified including, but not be limited to, the following:

- (1) Isolated Phase Bus Duct for Generator Output with accessories, consisting of:
 - a. Main Bus : Generator to generator circuit breaker, generator circuit breaker to generator transformer
 - b. Branch Bus : Isolated Phase Bus Connection to Unit Auxiliary Transformer
 - c. Tap Off Bus : Isolated Phase Bus Connection to Voltage Transformer including Surge Protection Cubicle, Excitation Power Transformer Cubicle (if required) and Static Frequency Converter (if required).
 - d. Neutral Bus : Generator neutral connection bus.

- e. Neutral Grounding
Bus Connection : Connection from the generator neutral to the Neutral Grounding Cubicle by cable.
 - f. Supporting Structures and foundation for above equipments
- (2) Six (6) Voltage Transformers (2 sets of 3 each) on generator side and three (3) voltage transformer (1 set of 3 each) on generator transformer side, including secondary fuses and generator surge protection with accessories.
- (3) One (1) Generator Neutral Grounding Cubicle with grounding transformer, grounding resistor, protection equipment and accessories with necessary connections.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards and the following specific standards.

2.1 National

PUIL : Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electrical Installation - Indonesia)

SPLN : Standard Perusahaan Umum Listrik Negara

2.2 International Electrotechnical Commission

IEC 60044 : Instrument transformers

IEC 60071 : Insulation co-ordination

IEC 60076 : Power transformers

IEC 60099 : Lightning arresters

IEC 60105 : Recommendation for commercial-purity aluminum bus bar material

IEC 60137 : Insulated bushings for alternating voltages above 1000V

IEC 60186 : Voltage transformers

- IEC 60273 : Dimensions of indoor and outdoor post insulators and post insulator units for systems with nominal voltages greater than 1000 V
- IEC 60529 : Degrees of protection provided by enclosures (IP Code)

3. OPERATING AND DESIGN CONDITIONS

3.1 General

The isolated phase bus duct shall be self-cooled heavy duty, metal clad and designed for indoor and outdoor service in a tropical climate along the seacoast.

3.2 Rating

3.2.1 Bus

The bus shall have the following rating:

- (1) Design voltage line to line : Not less than 5% above generator rated voltage
- (2) Operating voltage : Generator rated voltage
- (3) Rated frequency : 50 Hz
- (4) Rated continuous current :
 - a. Main Bus : Not less than 5% above generator current at maximum capability.
 - b. Branch Bus : Not less than 5% above the currents for maximum MVA of unit auxiliary transformers.
 - c. Tap Off Bus : Not less than 5% above the maximum equipment current requirements.
 - d. Neutral Bus : Not less than 5% above the maximum equipment current requirements.
 - e. Neutral Grounding Bus Connection : Not less than 5% above the maximum current equipment requirements.

(5) Rated momentary current (3-phase rms asymmetrical)

Rate momentary current shall be calculated by Contractor, giving not less than 20% margin. The Calculation shall be sent to the Owner for approval.

(6) Insulation level (minimum)

a. Rated impulse withstands voltage (peak) : as per generator voltage *)

b. Rated power-frequency short duration withstand: as per generator voltage *)

Note:*) Items to be determined by the Contractor and subjected to the Owner's Approval.

(7) The hottest spot temperature rise, at rated current, above ambient temperature of 40°C of air surrounding the bus enclosures and cubicles, shall not be more than :

a. Bus conductors and their silver surfaced, welded or brazed connections and disconnect links : 65°C

b. Generator and transformer connections as limited by the flexible bushings : 65°C

c. Bus enclosure and support structures where in accessible..... : 65°C

d. Bus enclosure and support structures where easily accessible : 30°C

e. Average temperature of the air surrounding outgoing insulated cables : 15°C

f. Connections to insulated cable (silver surfaced or equivalent) : 45°C

g. Cubicles accessible to personnel : 30°C

h. External surfaces of other non-current carrying parts : 70°C

4. DETAILED REQUIREMENTS

4.1 General

The bus shall be of the naturally-cooled isolated phase, metal enclosed, high current carrying capacity type. Each phase of the bus shall be fully air-insulated. Three (3) enclosures shall be mounted to form a completely self-supporting 3-phase unit assembly. The spacing between enclosures and the design and construction of the supporting structures shall be such as to

achieve the economy of operation, performance and reliability herein specified.

All bus conductors and connections, insulators, insulator supports, enclosures and supporting structures, when installed, shall have sufficient mechanical strength to ensure continuous and satisfactory operation under normal operating conditions and shall withstand, without incurring damage, the effect of any momentary current resulting from a three-phase, line-to-line or line-to-ground short circuit. The momentary current shall be as specified in Clause 3.2.1.

Outdoor portions of the assembled bus and structures shall be designed to withstand wind, beating rain and solar radiation as specified in this Specification without affecting the operation of the bus. The design shall be based on a minimum of 1100 W/m² of solar radiation. Sunshades shall not be used.

The isolated phase buses shall consist of rigid electrical conductors mounted on high quality porcelain or cast resin insulators in its own individual metal enclosure. Each phase shall be enclosed in a separate weather and dust tight non-magnetic metal enclosure, separated by an air space from the adjacent phase. The three phases shall be securely braced, mounted on and supported by a substantial metal structure. Where taps are specified from the bus to the cubicles housing auxiliary equipment, the tap connections and cubicles shall be of the isolated phase construction. Drain holes, screened to prevent entry of vermin, shall be provided to permit condensed moisture to escape.

The Contractor shall design, supply and install all necessary support structures, which shall be hot-dipped galvanized steel and shall provide details of the foundations and all penetrations that are required. The support structures shall be grounded.

Each section of the bus enclosure shall be permanently marked with phase identification colors. Red, yellow and black as per PUIL Standard shall be used.

4.2 Bus Conductor and Insulators

The bus conductor shall be either copper or high conductivity aluminum. Bus insulators and supports shall be braced or designed to prevent the permanent displacement of any part due to mechanical stresses resulting from short circuits.

Flexible connections, preferably braided, shall be provided in long straight runs and at all equipment terminations. Expansion connections shall be provided to eliminate the effects of thermal expansion, contraction and vibration stress transfer. Bolted flexible connections shall be silver plated before assembly.

Disconnecting links (or other suitable means) shall be provided to allow the isolation of the followings:

- (1) Primary of the Unit Auxiliary Transformer
- (2) The Generator
- (3) The Generator Circuit Breaker
- (4) Primary of Generator Transformer
- (5) Excitation Transformer (If required)
- (6) Starting Device Transformer (If applicable)

Isolation of the transformers and generator shall be possible using readily available hand tools. The bus system shall be able to meet all insulation tests with a link removed.

4.3 Bus Enclosures

The bus enclosures shall be of the continuous welded type and therefore designed to minimize the surrounding magnetic field and reduce the effect of induced currents in the adjacent structures and equipment. They shall be fabricated of non-magnetic material.

The connection between the bus duct enclosures and flanges at the transformer tank shall be provided with insulated gaskets to prevent circulating currents in the transformer tank top caused by induced voltages in the bus duct enclosures.

Suitable barriers shall be provided within the enclosures to seal-off sections where the bus enclosure passes through the power plant exterior building wall. Flexible joints or expansion joints shall be furnished at enclosure terminations for a vibration free connection.

Filter drains shall be provided at terminations to equipments, at building walls, floors, or roof and at all low points in the horizontal portions of the bus structure, to ensure the elimination of any condensation that might collect in the enclosures.

4.4 Supporting Structures

Rigid metal frames shall be furnished to support the three-phase sections of the bus structure. The Contractor shall furnish all supporting structures, including columns, base plates, hangers, anchor bolts and all necessary hardware.

All steel structures shall be hot-dipped galvanized. Where necessary to avoid excessive heating, the framework supporting the bus structure shall be of nonmagnetic materials, or it shall be provided with low resistance shielding bands or short circuiting rings.

4.5 Ground Bus

The enclosure on each phase of the Isolated Phase Bus shall also function as a ground bus and shall be capable of carrying the rated momentary current (rms asymmetrical) of the bus for a period of at least two (2) seconds.

The Contractor shall furnish a suitable size of copper or equivalent ground bus in each auxiliary equipment cubicle. The cubicle structures and the bases and frames of all items of electrical equipment shall be connected to the grounding bus. All ground buses shall have provision for connections to the station grounding system and/or building steel.

4.6 Painting

Paint shall be suitable for the service and the environmental conditions. Finish paint colors will be specified by the Owner later.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 General

The Contractor may submit test certificates of type tests which have been carried out on identical equipment. The type tests prescribed shall be carried out at the Contractor's expense in all cases where such certificates are not available or rejected.

5.2 Shop Tests

The following shop tests shall be performed by the Contractor:

- (1) One (1) sample of each part shall have type test certificate in accordance with relevant IEC or equivalent and shall include the following:
 - a. Impulse withstand test
 - b. Temperature rise test
 - c. Short circuit test
 - d. Joint conductivity test
 - e. Impulse tests
 - f. Withstand high voltage tests

If records exist for these tests on comparable equipment, they may be accepted in place of further type tests.
- (2) Tests on voltage transformers and grounding transformer shall be in accordance to IEC 60186 and IEC 60076, respectively. Test certificates and excitation curves shall be submitted for each transformer.
- (3) Isolated phase bus ducts shall be subjected to routine tests as specified in relevant IEC standard or equivalent and shall include at least the following:
 - a. Visual and dimensional checks
 - b. Radiographic weld examination
 - c. Joint conductivity test

5.3 Site Tests

The following site tests also described in Section 4 shall be performed to ensure a satisfactory system. The tests shall include, but not be limited to, the followings:

- (1) Insulation resistance test of all high voltage and all low voltage sides.
- (2) High voltage tests.
- (3) Resistance measurements of bus conductors.
- (4) Welding check and resistance measurements of welded joints in metal enclosed bus.
- (5) Load tests to measure temperature rise and to locate any hot spots in the enclosure or supporting steel structures.
- (6) Functional testing of all equipment.
- (7) Air pressure testing.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for the isolated phase bus duct shall conform to the requirements included in Part 4, Section 2 Subsection 2.7 and shall include, but not be limited to, the following:

- (1) Welding
- (2) Nondestructive Testing
- (3) Packaging and Shipping
- (4) Shop Test Procedures
- (5) Cleaning and Painting

The documentation as specified shall include but not limited to the following:

- (1) Material certificates
- (2) All shop test reports
- (3) All other documentation required by applicable codes and standards.

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